OpenCloning: an Open Source alternative to SnapGene and Benchling that supports automation and integration with research software

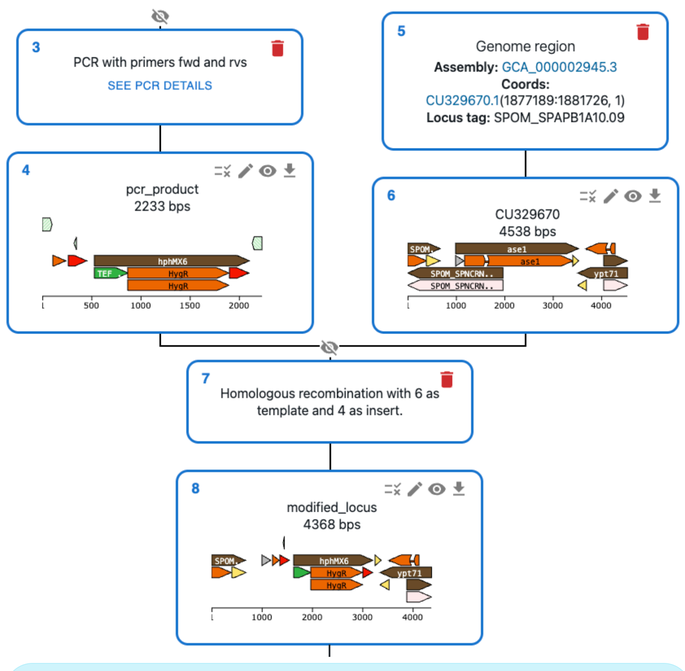
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What is OpenCloning?

An **open-source platform to plan, document and automate DNA engineering**.

- **Design in the browser or script it in Python**.
- **Every step is recorded** in an open data model: every construct is traceable and reproducible.
- **Download entire workflow** as JSON.
- **Free online**, and can be **self-hosted**.



Use it at https://app.opencloning.org

Automate it end-to-end

- Built on **pydna**: every method available in **Python**.
- Automate your design with scripts, and turn them into **reusable robotic workflows** for lab automation.

```
product, *_ = golden_gate_assembly(
[insert1, insert2, vector], [BsaI])
print(product.history())

└─ product (Dseqrecord(o39))
    └─ RestrictionAndLigationSource
        ├─ insert1 (Dseqrecord(-27))
        ├─ insert2 (Dseqrecord(-27))
        └─ vector (Dseqrecord(o35))
```

Example Notebook

OpenCloning on GitHub

Integrates with your lab software

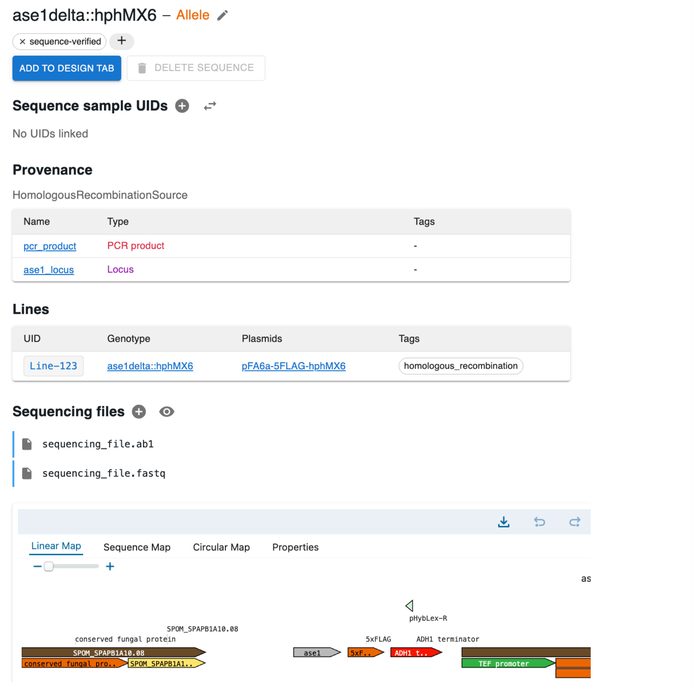
- API enables **easy integration** with ELNs, LIMS and lab equipment.
- Already integrated with **eLabFTW**.

eLabFTW

OpenCloningDB

Go beyond cloning and manage your collection

- **Sequences, oligos, strains and sequencing data** in one database, connected by provenance.
- **We can host it for your lab**, or you can **self-host it**: it's open source.



Try OpenCloningDB in your lab

We'll help you set it up. Your feedback shapes what we build next.

Get in touch

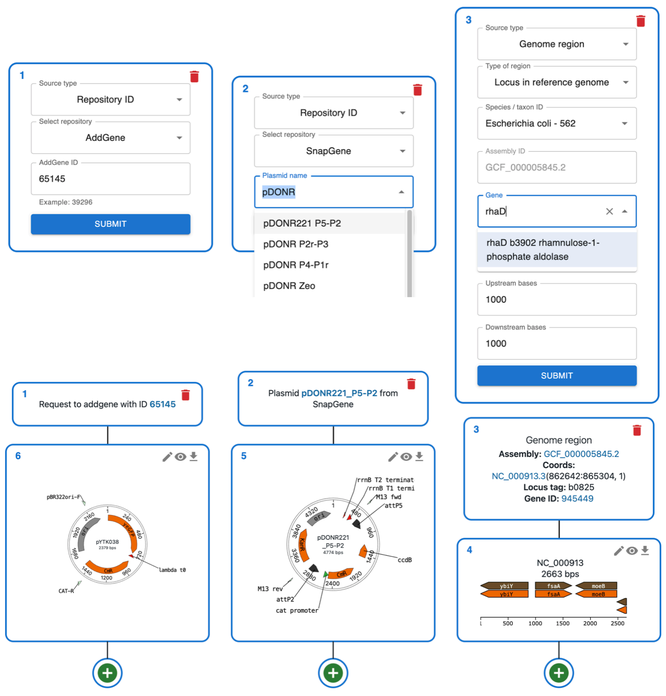
Stay updated

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The web interface

Import from anywhere

- Local files
- AddGene
- GenBank
- NCBI genomes
- SnapGene website
- Benchling link
- EuroScarf
- ELNs



Best Golden Gate tool

- **Faster and simpler than Benchling and SnapGene.**
- **Define your assembly syntax once** (the overhangs of each part type).
- **Plasmids are sorted automatically** into the right category (promoter, CDS, terminator...).
- **Pick parts from dropdowns** and get every combination, fully documented.
- **Automatic domestication** coming soon.



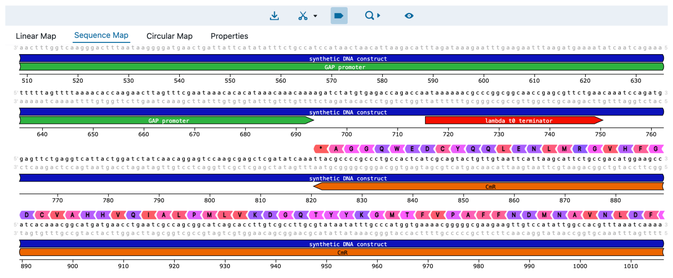
Create your own syntax at https://syntax.opencloning.org/

Techniques supported

- Golden Gate
- Gibson Assembly
- Restriction - ligation
- PCR
- Hom. Recombination
- CRISPR-HDR
- Overlap extension PCR
- Gateway™
- Polymerase extension
- Oligo hybridization
- *In vivo* assembly
- Cre-loxP
- Recombinases



Rich feature viewer



Primer design

- Gibson Assembly
- Hom. Recombination
- CRISPR-HDR
- Restriction - ligation
- Checking PCR
- Gateway™

